Lancaster University

2018 EXAMINATIONS

PART II – Third Year

MATHEMATICS & STATISTICS

Math 336: Multivariate Statistics in Machine Learning $1\frac{1}{2}$ **hours**

You should answer ALL SECTION A questions and ONE SECTION B question. If you answer more than one question from Section B then the best question will count. There is a total of 70 marks. You may only use the calculator that has been provided.

Questions start over the page

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SECTION A

A1. (a) Define the following concepts in the context of VC-analysis. [6]

i. Shattering

ii. Growth function

iii. VC-dimension

(b) We have a set of 5 points $\{x_1, \ldots, x_5\}$. It is known that a hypothesis set $\mathcal{H}$ implements a total of $M = 30$ dichotomies on these 5 points. Can we conclude that 5 is a break-point for $\mathcal{H}$? Justify your answer. [4]

(c) Let $\mathcal{H}$ be a hypothesis set. It is known that $\mathcal{H}$ implements $M < 16$ dichotomies for every set of 4 points $x_1, \ldots, x_4$. Can we conclude that 5 is a break-point for $\mathcal{H}$? Justify your answer. [4]

A2. Alice would like to predict her academic performance (pass/fail) based on the number of hours per week that she spends at the library, denoted by X taking value in $\mathbb{R}^+$. Let $Y = 1$ denote pass and $Y = 0$ denote failure.

(a) Assume that the conditional probability of pass $\mathsf{P}(Y = 1|X)$ is given by $\eta(X) = \frac{X}{X+3}$.

i. Give the Bayes-optimal classification rule. [5]

ii. Assuming that X is uniformly distributed between $[0, 12]$ what is the loss $L(g^*)$ associated with the Bayes-optimal classifier g^* in this case? [5]

(b) Suppose that Alice does not have access to the conditional probability of pass $\mathsf{P}(Y = 1|X)$. Instead, she is given a historical dataset $\{(x_1, y_1), \ldots, (x_N, y_N)\} \subset \mathbb{R} \times \{0, 1\}$ corresponding to $N = 10000$ students, where for each $i \in 1..N$, x_i denotes the average amount of time per week that Student i spends at the library and y_i denotes whether or not Student i has passed. Alice decides to take a Machine Learning approach to approximate $\mathsf{P}(Y = 1|X)$ using the given dataset. Since the number of data points is large, she is expecting to be able to provide a hypothesis that will definitely approximate $P(Y = 1|X)$ well on new, unseen, data points. Is this expectation correct? If not, what is the correct expectation? Justify your answer. [4]

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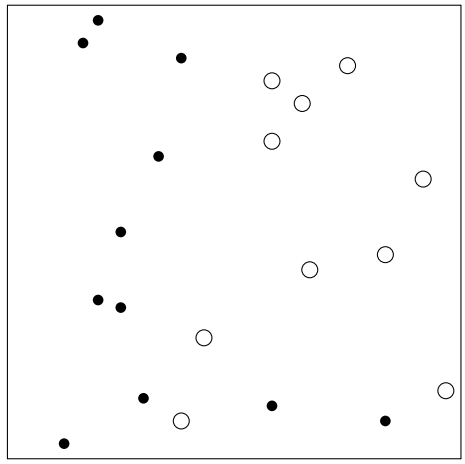
SECTION A continued

A3. Consider a binary classification problem with training set $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)$, where $\mathbf{x}_i \in \mathbb{R}^2$ and $y_i \in \{+1, -1\}$ for $i \in 1..N$. A sample plot of the training data is given below, where $\bullet$ denotes positive training examples (with label $+1$) and $\circ$ denotes negative training examples (with label -1).



(a) Is this is a supervised or an unsupervised learning problem? Justify your answer. [1]

(b) Which one of the following algorithms would you recommend for this classification problem? Justify your answer. [1]

- Perceptron Learning Algorithm
- Linear SVM
- k-means
- kNN

(c) Give a step-by-step description for the algorithm that you have selected in Part (b) above. [3]

(d) Suppose that the classification is done via an algorithm that uses a finite hypothesis class $\mathcal{H}$ with $|\mathcal{H}| = 1024$. Show that $\mathcal{H}$ has VC-dimension $d_{VC}(\mathcal{H}) = 10$. [4]

(e) Give an example of a hypothesis class with infinite VC-dimension. Justify you answer. [3]

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SECTION B

B1. (a) We wish to select between two hypothesis classes, $\mathcal{H}_1$ and $\mathcal{H}_2$ in order to choose a classifier to train. We have access to a small number of $N < 100$ data points.

i. Describe an appropriate method to reliably choose the correct hypothesis class, justifying your choice. [4]

ii. Assuming that $\mathcal{H}_1$ is a hypothesis class of fixed complexity, draw a graph showing two curves: test-error vs. the number of training samples N and training error vs. the number of training samples N. [4]

(b) Consider a hypothesis class $\mathcal{H}$ with VC-dimension $d_{VC}(\mathcal{H}) = d$ and growth function $m_{\mathcal{H}}(N),\ N \in \mathbb{N}$. Show that $m_{\mathcal{H}}(N) \leq N^d + 1$ for all $N \in \mathbb{N}$. [**Hint:** You may assume without justification that if $m_{\mathcal{H}}(k) < 2^k$ for some $k \in \mathbb{N}$, then for all $N \in \mathbb{N}$ we have $m_{\mathcal{H}}(N) \leq \sum_{i=0}^{k-1} \binom{N}{i}$.] [6]

(c) Let $\mathcal{H}$ denote the hypothesis class corresponding to Perceptrons in $\mathbb{R}^d$. Show that the VC-dimension of $\mathcal{H}$ is $d_{VC}(\mathcal{H}) = d + 1$. [8]

(d) Let $\mathcal{H}$ denote the set of all circles on the plane $\mathbb{R}^2$. Show that $d_{VC}(\mathcal{H}) = 3$. [6]

(e) State whether or not the following statement is true. Justify your answer. [2]

A hypothesis set $\mathcal{H}$ with VC-dimension $d_{VC}(\mathcal{H}) \geq d$ can shatter any set of d points.

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SECTION B continued

B2. (a) For a fixed $n \in \mathbb{N}$, and a set $S \subset \{1, \ldots, n\}$ we define a *parity function* as

$$f_S : \{0,1\}^n \to \{0,1\}$$
$$f_S(\mathbf{x}) = \left(\sum_{i \in S} x_i \right) \bmod 2.$$

For example, if $n = 5$, $\mathbf{x} = 01011$ and $S = \{2, 4\}$ then $f_S(\mathbf{x}) = 0$. Consider a class $\mathcal{H} := \{f_S : S \subset \{1, \ldots, n\}\}$ of parity functions. Show that $d_{VC}(\mathcal{H}) = n$. [6]

(b) The Perceptron Learning Algorithm (PLA) is applied to the following linearly separable data set $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, where $\mathbf{x}_i \in \mathbb{R}^2$ and $y_i \in \{+1, -1\}$ for $i \in 1..N$. Let $\mathbf{w}^* \in \mathbb{R}^2$ be an optimal weight vector, and denote by $\mathbf{w}(t)$ the PLA weight vector at iteration t, with $\mathbf{w}(0) = \mathbf{0}$. *In answering the questions below, you may assume validity of the previous statements. For example, in answering Part(iii) you may assume that the statements in Part(ii) and Part(i) are correct.*

(i) Let $\rho := \min_{1 \le n \le N} y_n \mathbf{w}^{*T} \mathbf{x}_n$. Show that $\rho > 0$. [2]

(ii) Show that $\mathbf{w}(t)^T \mathbf{w}^* \ge \mathbf{w}^T(t-1)\mathbf{w}^* + \rho$ and conclude that $\mathbf{w}(t)^T \mathbf{w}^* \ge t\rho$.
[**Hint:** You may assume that $\rho > 0$ where ρ is defined in Part (a).] [5]

(iii) Show that $||\mathbf{w}(t)||^2 \le ||\mathbf{w}(t-1)||^2 + ||\mathbf{x}(t-1)||^2$. [6]

(iv) Show that $||\mathbf{w}(t)||^2 \le tR^2$ where $R = \max_{1 \le n \le N} ||\mathbf{x}_n||$. [5]

(v) Using the above steps show that

$$\frac{\mathbf{w}(t)^T}{||\mathbf{w}(t)^T||} \mathbf{w}^* \ge \sqrt{t} \frac{\rho}{R}$$

and hence prove that $t \le \frac{R^2 ||\mathbf{w}^*||^2}{\rho^2}$. [6]

`end of exam`